Rahul Yedida

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Website :: GitHub :: LinkedIn :: Google Scholar :: Google Developers

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Education

North Carolina State University — Raleigh, NC

Ph.D. Computer Science — *Aug 2019 - Jul 2024*

- Advisor: *Dr. Tim Menzies*
- Dissertation: *Guidelines for the Application of Neural Technologies in Software Analytics (or: How to Do More with Less in SE)*

PES University — Bangalore, India

B.E. Computer Science – Advisor: Dr. Snehanshu Saha — *Aug 2015 - May 2019*

Employment

LexisNexis Legal & Professional — Raleigh, NC

Senior Data Scientist II — *Oct 2026 - Present*

- Fine-tuned small structured-decision models to match LLM-baseline performance (macro-F1 92.7%) at 177 ms median latency on-device, after benchmarking 15 models across 6 tasks.

Senior Data Scientist I — *May 2024 - Oct 2026*

- Technical lead for an agentic legal drafting system; led its migration from a RAG pipeline and own its ongoing development.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Built a contrastive-learning filter using InfoNCE with alignment and uniformity losses to remove unrelated legal passages from search results; median filtering latency of 12 ms across 40 queries.
- Built team developer tooling: custom Python lint rules beyond `ruff`'s set, enforced in CI; a Rust MITM proxy with request replay that teammates use to debug without waiting on upstream responses; and a web UI for visualizing intermediate agent steps.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

North Carolina State University — Raleigh, NC

PhD Student — *Aug 2019 - Jul 2024*

- **State-of-the-art hyper-parameter optimization:** Derived loss-curvature estimates for smoothness- and flatness-guided search; the smoothness-based method matched or beat prior SOTA on 17 of 19 software datasets.
- **Better, faster deep learning for SE:** Introduced fuzzy oversampling and optimized neural-network hyper-parameters, surpassing prior SOTA with median relative gains of 15% in defect-prediction F1 and 12% in AUC across four code smells. Built tuned feedforward models that matched or beat SOTA across 15 issue-lifetime benchmarks in minutes instead of hours.
- **Microservice partitioning:** Developed a graph neural network partitioner with a custom weighted loss; reduced the share of cross-service calls by a median 66% versus prior SOTA across four applications.
- **Semi-supervised learning:** Combined oversampling, nearest-neighbor labeling, and hyper-parameter optimization for state-of-the-art static-analysis warning triage: median AUC 0.92 on eight Java projects using 10% of training labels.
- **Teaching:** TA for 830 students over five semesters in C programming and software engineering.

Amazon — New York, NY / Bellevue, WA

Software Dev Engineer Intern — *May 2023 - Aug 2023*

- Implemented and shipped profile locks for Prime Video on Echo Show devices.
- **Technology:** React Native, TypeScript

Software Dev Engineer Intern — *May 2022 - Jul 2022*

- Developed a full-stack system to publish announcements in scorecards used by delivery service partners.
- **Technology:** React/Redux, TypeScript, Redux Saga, DynamoDB, Java Spring

Recent Publications

See full list on Google Scholar.

1. Nagarsekar, A., Bhat, M. P. A., Nesarkar, A., Godhwani, V., **Yedida, R.**, Challa, A., Danda, S., & Saha, S. (2026). A Generalisation Signal Need Not Be a Model-Selection Signal. In *I Can't Believe It's Not Better: Failure Modes of AI in Biology Workshop (NeurIPS 2026)*.
2. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering, 51(6)*.
3. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM, 66(2), 75-81*.
4. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering, 49(4), 2856-2872*.
5. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications, 219, 119673*.
6. **Yedida, R.**, & Menzies, T. (2022). How to Improve Deep Learning for Software Analytics (a case study with code smell detection). In *2022 IEEE/ACM 19th International Conference on Mining Software Repositories (MSR)*.
7. **Yedida, R.**, & Menzies, T. (2021). On the Value of Oversampling for Deep Learning in Software Defect Prediction. *IEEE Transactions on Software Engineering, 48(8), 3103-3116*.
8. Agrawal, A., Yang, X., Agrawal, R., **Yedida, R.**, Shen, X., & Menzies, T. (2021). Simpler Hyperparameter Optimization for Software Analytics: Why, How, When?. *IEEE Transactions on Software Engineering, 48(8), 2939-2954*.
9. Yang, X., Chen, J., **Yedida, R.**, Yu, Z., & Menzies, T. (2021). Learning to recognize actionable static code warnings (is intrinsically easy). *Empirical Software Engineering, 26(3), 1-24*.
10. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2021). Lessons learned from hyper-parameter tuning for microservice candidate identification. *Proceedings of the thirty-sixth IEEE/ACM International Conference on Automated Software Engineering (ASE)*.
11. **Yedida, R.**, & Saha, S. (2021). Beginning with Machine Learning: A Comprehensive Primer. *The European Physical Journal Special Topics*, 230(10), 2363-2444.
12. Saha, S., Nagaraj, N., Mathur, A., **Yedida, R.**, & Sneha, H. R. (2020). Evolution of novel activation functions in neural network training for astronomy data: habitability classification of exoplanets. *The European Physical Journal Special Topics, 229(16), 2629-2738*.
13. **Yedida, R.**, Saha, S., & Prashanth, T. (2020). LipschitzLR: Using theoretically computed adaptive learning rates for fast convergence. *Applied Intelligence, 1-19*.
14. Sridhar, S., Saha, S., Shaikh, A., **Yedida, R.**, & Saha, S. (2020, July). Parsimonious Computing: A Minority Training Regime for Effective Prediction in Large Microarray Expression Data Sets. In *2020 International Joint Conference on Neural Networks (IJCNN) (pp. 1-8)*.
15. Khaidem, L., **Yedida, R.**, & Theophilus, A. J. (2019, November). Optimizing Inter-nationality of Journals: A Classical Gradient Approach Revisited via Swarm Intelligence. In *International Conference on Modeling, Machine Learning and Astronomy (pp. 3-14). Springer, Singapore*.

Preprints

1. Hearn, T., Evans, W., **Yedida, R.**, Abdelshiheed, M., Rebei, A., & Khazaeli, S. (2025). Automated Legal Brief Drafting via Iterative Retrieval-Augmented Generation.
2. Rashid, P., Pal, S., Samudrala, V. K., **Yedida, R.**, & Gehringer, E. (2025). Enhancing Peer Assessment Reliability: A Probabilistic Approach Integrating Formative and Summative Feedback.
3. Mehendale, S., Challa, A., **Yedida, R.**, Danda, S., Sarkar, S., & Saha, S. (2025). Radon–Nikodym Derivative: Re-imagining Anomaly Detection from a Measure Theoretic Perspective.
4. **Yedida, R.**, & Saha, S. (2024). Strong convexity-guided hyper-parameter optimization for flatter losses.

Funding

$5,000, Google Cloud Academic Research Grant, Feb 2022

Service to Profession

Contributing Member, Python Software Foundation (Mar 2021 - Present)

Guest Editor, IEEE Software SI: The Impact of AI on Productivity and Code 2025; ASEj SI: Replications and Negative Results (RENE) 2025; EMSE SI: Replications and Negative Results (RENE) 2025

Workshop Co-Chair, Intelligent Software Engineering (ASE '25, ICST '26); Replications and Negative Results (ASE '24)

PC Member, AAAI 2025-2027; FSE 2026-2027; ICSE 2026-2027; EDM 2026; ASE Industry Track 2026; AI Foundation Models and Software Engineering (FORGE) 2024; ICSME Artifact Evaluation Track, 2021-2023; ASE Artifact Evaluation Track, 2022; International Conference on Modeling, Machine Learning, and Astronomy (MMLA), 2019

Reviewer, TOSEM 2026; TSE 2025-2026; NeurIPS 2023, 2025; EMSE 2021, 2025; ASEj 2025; ICML 2024-2025; ICLR 2024-2025; NCAA 2023-2025; TMLR 2024; Neural Processing Letters 2023-2024; Artificial Intelligence Review 2023; J. Big Data, 2023; ASE 2023; IEEE SSCI 2020

Honors

Distinguished Reviewer, ICSE 2026 Mar 2026

Google Developer Expert - ML and GCP Mar 2025 - Present

A global network of experts recognized by Google for deep technical expertise, community leadership, and mentorship.

Google Cloud Champion Innovator - Cloud AI/ML Oct 2022 - Mar 2025

A Googler-nominated global network of 500+ technical experts in Google Cloud; merged into the Google Developer Experts program in 2025.

Google Cloud Research Innovator Feb 2022

Research Innovators are a global community of researchers driving scientific breakthroughs with Google Cloud.

Invited Talks

Feb 2024, "*Improving deep learning performance using theoretical ML*" at BITS Pilani, KK Birla Goa Campus, India

Relevant Projects

pi-permission-system (maintained fork) May 2026 - Present

TypeScript, Node.js *GitHub*

Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

Q&A System for Zotero Oct 2024 - Present

Rust, SQLite, LanceDB *GitHub :: crates.io*

Built an academic question-answering and semantic search system over a Zotero library. Developed a Rust RAG library with configurable LLM, embedding, and reranking providers, tool calling, and limits on concurrent embedding requests. Wrote a custom PDF parser for academic papers (median throughput ≈ 2,400 pages/s across 11 PDFs; M4 Pro).

JournalBear Jan 2017 - Jul 2026

TypeScript, React, Zustand, Electron, SwiftUI *GitHub :: App Store :: Softpedia*

Built and shipped a journaling app with AES-256 encryption, an Electron/React desktop client, and native SwiftUI apps for iOS and macOS.

RAISE Aug 2020 - Jun 2026

Python, Keras, PyTorch *GitHub :: PyPI*

Sole developer of a Python package for reproducible ML experiments, providing data loading, evaluation metrics, statistical tests, and reference implementations of research algorithms. Uses Keras with a PyTorch backend; tested on Python 3.11-3.14 across Linux, Windows, and Apple Silicon macOS. Used by multiple research groups; downloaded 79k+ times on PyPI.

Flatness-guided HPO Nov 2023 - May 2026

Python, PyTorch, Keras *GitHub :: Paper*

Developed a hyper-parameter optimizer using closed-form loss-curvature estimates to select configurations with flatter losses. Implemented comparative benchmarks on MNIST, SVHN, OpenML, and Bayesmark.

The 1 Billion Row Challenge (1BRC) Feb 2026 - Apr 2026

Zig *GitHub*

Wrote a Zig implementation for the 1BRC, running in 1.20s (M4 Pro, 14 core) using mmap, open addressing (Wyhash + Robin Hood probing), and packed SIMD.

Programmable Resumes Aug 2023 - Mar 2025
Python *GitHub*
Developed a resume specification language and Python compiler that generates PDFs from structured resume data via LaTeX. Supports tag-based role filtering, two LaTeX templates, and parallel builds. Wrote this resume using this tool.

pysh May 2021 - Dec 2024
C++, TMLanguage *GitHub*
Developed a superset of Python that allows running shell code natively, with a VS Code syntax highlighting extension.

GHOST Jun 2020 - Mar 2022
Python, Keras *GitHub :: Paper*
Developed the GHOST reference implementation for software defect prediction, combining adaptive oversampling with DODGE hyper-parameter optimization. Implemented configurable ablation experiments and Scott-Knott statistical comparisons using RAISE.

Threaded Discussions Website Feb 2021 - Jun 2021
MongoDB, Node.js, React *GitHub*
Built a companion site for video calls supporting Reddit-style threaded discussions.

NearConnect Nov 2020 - Mar 2021
iOS, SwiftUI *GitHub :: App Store*
iOS app to connect with people nearby using multicast peer-to-peer connections.

Results Scraper Mar 2018 - Aug 2018
MongoDB, Express.js, React, Node.js *GitHub*
Website for scraping university examination results and displaying charts and printable reports, with caching.

Video Sharing Website Oct 2017 - Dec 2017
MySQL, Express.js, React, Node.js, Sass, Elasticsearch *GitHub*
Simplified implementation of a video-sharing website with subscriptions and custom searching.

Web development projects Jan 2017 - Nov 2017
MongoDB, Express.js, React, Node.js, Sass, D3.js *CodePen*
Projects include URL shortener, rogue-like dungeon crawler game, voting application, Simon game, land surface temperature heatmap, and mapping meteorite impacts across the globe.

Xtreme Calculations Apr 2013 - Oct 2017
VB.NET, Python *Softpedia*
Windows math software to solve scientific and mathematical problems, with over 30k downloads across multiple sites.

Video Indexer Jun 2017 - Sep 2017
C++, Qt, Sphinx *GitHub*
Cross-platform desktop application to detect the time(s) a given keyword was spoken in a video.

Skills

Languages: Rust, Python, TypeScript, Java, C++, Gleam

ML/AI: PyTorch, Keras, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK

Frameworks: React, FastAPI, Litestar, FastAPI

Databases: LanceDB, SQL, MongoDB, DynamoDB

Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)